

SURAJ DIAS

 SurajDias

 theSurajDias

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ABOUT

Computer Science and Business Systems Engineering student with a strong interest in backend engineering, compiler design, and machine learning, experienced in building scalable full-stack applications.

SKILLS

Languages: Java, Python, C, JavaScript, TypeScript

Frameworks: React.js, FastAPI, Flask, Node.js

Databases: MySQL, MongoDB

Tools: Git, Linux, Postman

PROJECTS

MiniCompiler Studio *React.js, Flask, Python*

Feb-Apr 2026

- Built a compiler visualization platform featuring a hand-written recursive-descent parser, AST generation, and a 5-pass optimizer (constant folding, algebraic simplification, copy propagation, dead-code elimination) that iterates to a fixed point over three-address code, securing Third Place in class-level Project Based Learning.
- Implemented a from-scratch regex engine that converts regular expressions into ϵ -NFAs using Thompson's construction, transforms them into minimized DFAs via subset construction and partition refinement, and visualizes each stage with animated simulations.

Career Guardian AI *React.js, FastAPI, Python*

Mar 2026

- Built an end-to-end resume analysis pipeline that parses uploaded PDFs, extracts skills using boundary-aware regular expressions, computes an ATS-style keyword coverage score, and generates actionable recommendations for resume improvement.
- Designed an interactive career roadmap that models missing skills as a filterable dependency graph with a deterministic layout, surfacing curated learning resources for each identified skill gap.

AutoOps Pro *React.js, TypeScript, FastAPI, Python*

Apr 2026 – Present

- Built a hybrid anomaly detection engine combining a 100-tree Isolation Forest with weighted threshold rules, fusing unsupervised and rule-based scores for explainable system health monitoring.
- Designed a weighted root-cause analysis engine that normalizes CPU, memory, latency, and error-rate contributions across incompatible units, alongside a what-if simulator that evaluates six remediation strategies using real-time trend analytics.

Customer Churn Intelligence Dashboard *Python, Streamlit, scikit-learn*

Sep-Nov 2025

- Built a customer churn prediction pipeline on telecom data, addressing class imbalance through oversampling and using a sigmoid-calibrated Random Forest to generate probability-based churn risk scores.
- Designed a feature-schema persistence system that preserves one-hot encoded feature alignment between training and inference, alongside a batch-scoring workflow supporting CSV and PDF report generation.

EDUCATION

St. Joseph Engineering College, Mangaluru

2023 – Present

B.E. Computer Science and Business Systems | CGPA: 7.82

CERTIFICATIONS

- Cisco – Introduction to Cybersecurity
- Generative AI with Large Language Models – Coursera
- Data Science Math Skills – Coursera